MIL-L-19891B
15 FEBRUARY 1963
SUPERSEDING
MIL-L-19891A
23 MAY 1960

MILITARY SPECIFICATION

LIGHT: ANGLE OF APPROACH, TYPE AA-1

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers a portable aviation light with an optical system which projects a tricolor beam. It is designated type AA-1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-515 — Plywood, Container Grade.

TT-P-636 — Primer Coating, Synthetic, Wood and Ferrous Metal.

TT-P-664 — Primer Coating, Synthetic Rust-Inhibiting, Lacquer-Resisting.

PPP-B-585 — Boxes, Wood, Wirebound.

PPP-B-591 — Boxes, Fiberboard, Wood-Cleated.

PPP-B-601 — Boxes, Wood, Cleated-Plywood.

PPP-B-621 — Boxes, Wood, Nailed and Lock-Corner.

PPP-B-636 — Boxes, Fiberboard.

PPP-T-60 — Tape, Pressure-Sensitive Adhesive Waterproof — for Packaging and Sealing.

MILITARY

MIL-B-18 — Batteries, Dry.

MIL-P-116 — Preservation, Method of.

MIL-R-900 — Rubber Sheets; and Cut, Molded and Extruded Special Shaped Sections — Synthetic, Medium Soft, Low Temperature, Gasket Application.

MIL-S-5002 — Surface Treatments (Except Priming and Painting) for Metal and Metal Parts in Aircraft).

MIL-A-5092 — Adhesive, Rubber (Synthetic and Reclaimed Rubber Base).

MIL-E-5272 — Environmental Testing, Aeronautical and Associated Equipment, General Specification for.

MIL-E-5556 — Enamel, Camouflage, Quick Drying.

FSC 6210

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:50Z
Check the source to verify that this is the current version before use.

MIL-M-7911 — Marking, Identification of Aeronautical Equipment, Assemblies and Parts.

MIL-P-8585 — Primer Coating, Zinc Chromate, Low-Moisture-Sensitivity.

MIL-B-10377 — Box Wood, Cleated, Veneer, Paper Overlaid.

MIL-L-10547 — Liners, Case, Waterproofing.

MIL-C-15328 — Coating, Pretreatment (Formula No. 117 for Metals).

MIL-C-25050 — Colors, Aeronautical Lights and Lighting Equipment, General Requirements for.

MIL-D-70327 — Drawings, Engineering and Associated Lists.

STANDARDS

FEDERAL

FED-STD-595 — Colors.

MILITARY

MIL-STD-129 — Marking for Shipment and Storage.

MIL-STD-130 — Identification Marking of U. S. Military Property.

DRAWINGS

SIGNAL CORPS ENGINEERING LABORATORIES

SC-D-20648 — Catch (Small) Assembly and Details.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CONSOLIDATED CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies of the above publication should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois.)

3. REQUIREMENTS

3.1 Preproduction samples. A preproduction sample ordered in accordance with the requirements of this specification shall be submitted for tests and written approval by the procuring activity prior to fabrication of any production equipment.

3.2 Materials. Materials shall conform to applicable specifications and shall be as specified herein. Materials which are not covered by applicable specifications or which are not specifically described herein shall be of the best quality, the lightest practicable weight, and suitable for the purpose intended.

3.2.1 *Selection of materials.* Specifications and standards for all materials, parts and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be governed by the requirements of the procuring activity or as specified in the contract, except as provided in the following paragraph.

3.2.2 *Standard parts.* Standard parts shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc., may be used, provided they possess suitable properties and are replaceable by the standard parts without alteration, and provided the corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there is no suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.2.3 *Fungus-proof materials.* Materials that are nutrients for fungi shall not be

Source: https://assist.dla.mil -- Downloaded: 202610-19T06:50Z
Check the source to verify that this is the current version before use.

used where it is practical to avoid them. Where used and not hermetically sealed in, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.3 **Protective treatment.** When materials are used in the construction of the light that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.4 **Design and construction.** The light shall be so designed and constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service. The objective shall be to design a light having optimum optical and physical characteristics coupled with minimum size and weight for use under extreme temperature, humidity, rain, snow, and rough handling conditions encountered in front line evacuation operation. There shall be as few separate parts or components, especially those that may be lost or improperly assembled during adverse servicing and field use conditions, as feasible.

3.4.1 *Interchangeability.* All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Change in manufacturer's part numbers shall be governed by the drawing number requirements of Specification MIL-D-70327.

3.5 **Components.** The light shall consist of an optical system mounted in a case and cover which is equipped with necessary electrical connections, a source of energy, a dimmer, a switch, and a support.

3.5.1 *Source of energy.* The light shall be powered by two dry batteries No. BA-200/U conforming to Specification MIL-B-18. Rated voltage shall be 6 volts. Unless otherwise specified, lights shall be furnished without batteries.

3.5.2 *Case.* The case shall be drawn or welded steel construction, coated as specified in 3.9 after fabrication, or it may be constructed of noncorrosive material. The wall thickness shall be not less than 0.08 inch if steel is used. If noncorrosive material is used, its thickness shall be such as to provide substantially equivalent strength and rigidity of steel. The perimeter of the case forming the sealing edge for the cover shall be free of irregularities to assure a good seal with the cover gasket. The case shall be designed to accommodate two batteries of the maximum specified dimensions in the battery compartment (see 6.4).

3.5.2.1 *Latches.* The case shall be provided with a side latch (catch) assembly and a hinge or two side latches to secure the case cover in a raintight relationship with the perimeter of the case. Latches shall be in accordance with Signal Corps Drawing No. SC-D-20648, except that springs shall be noncorrosive.

3.5.2.2 *Cover.* The case cover shall be of the same material and finish as the case. It shall properly fit the case. It shall be provided with electrical, sighting, levelling, and optical means as specified herein.

3.5.2.3 *Gasket.* The cover shall be provided with a gasket complying with Specification MIL-R-900, class II, cemented securely in place with cement complying with Specification MIL-A-5092. As an alternate, the gasket may be cemented to the case instead of to the cover.

3.5.2.4 *Latch keepers.* The cover shall be provided with latch keepers of proper design

Source: https://assist.dla.mil -- Downloaded: 202610-19T06:50Z
Check the source to verify that this is the current version before use.

to provide a secure mechanical connection with the latches specified in 3.5.2.1.

3.5.3 *Optical system.* The optical system shall be mounted in the cover in such manner that the light transmitting opening in the cover shall be raintight. The light transmitting closure means shall be recessed in its holder to protect it from mechanical injury during usage and storage. When projected on a screen normal to the optical axis, the beam pattern shall approximate a square having a horizontal band of yellow, of green, and of red reading from the top downward. The light intensity shall be as uniform as feasible by best design practice throughout the specified area. With a PR-12 lamp operating at rated current, the intensity of the beam shall be at least 2 candles in the square from 8 degrees below the red-green interface to at least 8 degrees above the red-green interface and horizontally at least 8 degrees each side of the optical axis. The angular width (dihedral angle) of the green band measured vertically shall be 2.5 degrees ± 0.1 degrees. The vertical width of interface regions of uncertain color recognition shall not exceed 0.25 degree. The light shall be a fixed focus type. No specific focusing or aligning shall be required to obtain the specified intensity or angular coverage. A spare No. PR-12 lamp shall be mounted adjacent to the operating lamp. The colors of the filters shall be within the limits for aviation yellow, green, and red as defined in Specification MIL-C-25050.

3.5.3.1 The minimum acceptable transmission ratios for the red, yellow, and green filters shall conform to the requirements for type I, class B colors as defined in Specification MIL-C-25050.

3.5.4 *Switch.* A three-position toggle switch shall be provided. The switch shall have a rubber boot or other watertight seal and shall be properly labeled ON, OFF, and DIM. It shall be protected against accidental operation during usage and storage.

3.5.5 *Dimming resistor.* A resistor of suitable capacity shall be provided for dimming the light from 100 percent brightness to 30 percent ± 5 percent brightness when a new battery is used as the source of energy.

3.5.6 *Electrical connection.* All electrical connections shall be made electrically and mechanically secure and of noncorrosive materials. The battery contacts shall be so designed that proper connection of the battery will be made in any one of the four possible orientations of the battery.

3.5.7 *Angular adjustment.* Means shall be provided for readily setting the lower edge of the green beam at any desired angular elevation between 0 degree and plus 30 degrees within 0.5 degree accuracy under adverse conditions of visibility and weather. It may consist of a level mounted on a lever pivotally attached to the cover. Locking means for any desired angular setting shall be provided. The free end of the lever, if a lever is used, shall have an indicator to indicate the angle of elevation on a protractor scale mounted on the cover. The scale or lever shall have a zero adjustment to correct for wear and small mechanical distortions. An L-shaped level to indicate when the cover is level longitudinally and laterally shall also be mounted on the upper surface of the cover.

3.5.7.1 *Sight.* A sight shall be provided for positioning the green beam above obstructions. The sight shall provide for scanning transversely 10 degrees each side of the center of the green beam coincident with its lower interface without moving the light. The sight shall be mounted on the case cover. The horizontal fiducial line shall be located to make an angle of 3 degrees below the red-green interface. The sighting device shall be designed to permit sighting with the eye located at a point at least 15 inches above the sight.

3.5.8 *Support.* The light shall be provided with a sturdy support for holding it stable under severe rotor or propeller blast when the light is set on sod, ice, uneven ground and the like encountered during adverse operating conditions. The support shall have readily

Source: https://assist.dla.mil -- Downloaded: 202610-19T06:50Z
Check the source to verify that this is the current version before use.

adjustable means for leveling the case laterally at least ± 15 degrees and for holding it firmly in such position. The support shall also have readily adjustable means for tilting and securing the case longitudinally with respect to the optical axis so that the lower edge of the green beam can be adjusted at least from — 10 degrees to + 45 degrees with respect to normal level position of the base of the support. A portion of the support shall be in the form of a round pipe or rod beneath the center of gravity of the light in operating position. It shall be at least 2 inches long and between ½ and ⅞ inch diameter to be inserted and clamped to the top of a substantially vertical angle-iron stake. It shall be laterally pivotal at least ± 15 degrees from vertical for leveling. Such stake and clamp will not be furnished unless specified by the procuring activity.

3.5.8.1 *Alternate leveling and support structure.* Alternate means of accomplishing the specified functions of support, leveling and adjustment listed in 3.5.7 and 3.5.8 may be furnished, provided the detailed method is approved in writing by the procuring activity in advance of production.

3.6 Operating temperatures. The light, exclusive of the battery, shall operate satisfactorily at any ambient temperature from — 54° C to + 55° C in accordance with the applicable requirements of Specification MIL-E-5272.

3.7 Weight. The weight of the light, completely assembled with two 1½ pound batteries, shall not exceed 8½ pounds.

3.8 Cleaning. The light shall be thoroughly cleaned of loose, spattered or excess solder, metal chips and other foreign material, after final assembly. Burrs and sharp edges, as well as resin flash that may crumble, shall be removed.

3.9 Coating. The case bottom, cover, catches, and hood (if made of steel) shall be finished inside and out with an acid-, corrosion-, and salt-water-resisting coating. The cleaning and surface treatment shall be in accordance with Specification MIL-S-5002. One coat of wash primer in accordance with Specification MIL-C-15328 (0.2 - 0.3 mil thick), two coats of primer in accordance with Specification MIL-P-8585, TT-P-636, or TT-P-664 (0.6 - 0.8 mil thick) and one coat of orange-yellow enamel (1.0 mil thick), Color No. 33538 of Federal Standard No. 595, in accordance with Specification MIL-E-5556, shall be applied.

3.10 Identification of product.

3.10.1 *Nameplate.* A nameplate properly and legibly filled in with information in accordance with Specification MIL-M-7911 for Navy purchases only, and Standard MIL-STD-130, shall be provided.

3.11 Workmanship. The light, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, painting, riveting, machine-screw assemblies, and freedom of parts from burrs and sharp edges.

3.11.1 *Dimensions.* Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the light assembly, they shall be held or limited accordingly. The height of the light with support shall not exceed 10 inches in lowest operating position.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection responsibility. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:50Z
Check the source to verify that this is the current version before use.

the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 **Classification of tests.** The inspection and testing of the lights shall be classified as follows:

a. Preproduction testing .....See 4.3
b. Acceptance testing .....See 4.4

4.3 **Preproduction testing.**

4.3.1 *Sampling instructions.* The first complete light produced under each contract and fabricated from production tools shall be the preproduction test sample and shall be delivered for inspection and testing as directed by the procuring activity, together with a certification by the cognizant inspector that the materials used in its manufacture are in accordance with pertinent specifications. (See 6.5.)

4.3.2 *Preproduction tests.* Preproduction tests of the lights shall consist of all tests contained in this specification as described under Test Methods and shall be performed in the order listed.

4.4 **Acceptance testing.** Acceptance testing shall consist of the following:

a. Individual tests .....See 4.4.1
b. Sample testing .....See 4.4.2

4.4.1 *Individual tests.* Each light shall be subjected to the following tests as described under 4.5 except when the inspector, at his discretion in lieu of testing each light, shall choose at random from each lot of 100 lights, or fraction thereof, 5 samples for tests to determine conformance with all requirements of this specification. If any of the 5 samples fails to pass the tests, the inspector shall choose at random 5 more samples. If any one of these samples fails, the entire lot shall be rejected.

a. Material & Workmanship..See 4.5.1
b. Operation .....See 4.5.11
c. Optical Performance .....See 4.5.12

4.4.2 *Sample testing.*

4.4.2.1 *Lot.* A lot shall consist of lights manufactured under essentially the same conditions and submitted for inspection at substantially the same time.

4.4.2.2 *Sampling instructions.* One light shall be selected at random from each lot of one hundred or fraction thereof produced in a lot and shall be called the acceptance sample.

4.4.2.3 *Sampling tests.* Each acceptance sample shall be subjected to the impact test of 4.5.3 and the photometric test of 4.5.13 in addition to the individual tests of 4.4.1.

4.4.3 *Rejection and retest.* When the acceptance sample fails to meet the tests, acceptance of all lights in the lot shall be withheld until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.4.4 *Individual tests may continue.* For production reasons, individual tests may be continued pending the investigation of a sampling test failure. Final acceptance of the entire lot shall not be made until it is determined that the lot meets all the requirements of the specification.

4.4.5 *Defects in lights already accepted.* The investigation of a test failure could indicate that defects may exist in lights already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.5 **Test methods.**

4.5.1 *Material and workmanship.* Each light shall be examined to determine conformance

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:50Z
Check the source to verify that this is the current version before use.

with this specification with respect to material, workmanship, and general requirements.

4.5.2 *Vibration tests.* The preproduction sample shall be subjected to the vibration test 4.7.11 (Procedure XI.–) of MIL–E–5272A.

4.5.3 *Impact test.* The complete light with batteries shall be dropped from a height of two feet upon a hard, rigid surface, once upon the bottom and once upon each of four sides. Evidence of damage, except to the lamp, shall be cause for rejection.

4.5.4 *Rain test.* This test shall conform to 4.10 of MIL–E–5272A.

4.5.5 *High temperature.* Test in accordance with 4.1.1 of MIL–E–5272; except the chamber temperature shall be 55° C instead of 71° C.

4.5.6 *Low temperature.* Test in accordance with 4.2.1 of MIL–E–5272A.

4.5.7 *Humidity.* Test in accordance with 4.4.3, Procedure III of MIL–E–5272A.

4.5.8 *Fungus resistance.* Test in accordance with 4.8.1 of MIL–E–5272A.

4.5.9 *Salt spray.* To insure compliance with applicable portions of 3.8, 3.5.2.1, 3.9, tests shall be conducted to conform to 4.6 of MIL–E–5272A.

4.5.10 **Interchangeability.** Parts should be checked to comply with 3.4.1.

4.5.11 *Operation.* Each completed light shall be given a two cycle test at room temperature to determine that it will operate properly at the specified minimum and maximum battery voltages and that it will receive and properly hold No. BA–200/U batteries of both maximum and minimum dimensions permitted by Specification MIL–B–18.

4.5.12 *Optical performance.* Each light shall be tested to see that the horizontal fiducial line of the sight is located to make an angle of 3 degrees ± 1 degree below that red-green interface.

4.5.13 *Photometric.* Photometric tests shall be conducted to determine compliance with light distribution and color requirements of 3.5.3. All parts shall reach normal operating temperatures before any readings are taken.

4.5.14 *Inspection of the preservation, packaging, packing, and marking for shipment and storage.* Sample items or packs and the inspections of the preservation, packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of Section 5, or the documents specified therein.

5. PREPARATION FOR DELIVERY

5.1 **Packaging.** Packaging shall be Level A or C as specified (see 6.2).

5.1.1 *Level A.* Each light assembly in its case shall be preserved and packaged in accordance with method IIB of Specification MIL–P–116. The case may be used as the interior container of the method IIB package, provided the item is adequately cushioned. The required amount of desiccant and the humidity indicator shall be placed inside the case in such manner that the desiccant shall not come in direct contact with any bare metal surfaces or parts. All openings and joints of the case shall be sealed with pressure-sensitive tape conforming to type I or II, class I of Specification PPP–T–60.

5.1.2 *Level C.* Lights shall be packaged in accordance with commercial practice.

5.2 **Packing.** Packing shall be Level A, B, or C as specified (see 6.2). Shipping containers, insofar as possible, shall contain an identical number of articles, shall be of uniform size, and shall be designed to enclose the contents in a snug, tight-fitting manner. The gross weight of the exterior shipping container when packed shall not exceed 200 pounds.

5.2.1 *Level A.* Unless otherwise specified, the interior packages shall be packed in exterior shipping containers in accordance with

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:50Z
Check the source to verify that this is the current version before use.

Specification PPP-B-601 (overseas type) or PPP-B-621. Plywood, when used, shall be type I or II, class 2 of Specification NN-P-515. The container shall be case lined in accordance with Specification MIL-L-10547. Strapping and closures shall be in accordance with the appendix to the applicable container specification.

5.2.2 *Level B.* Interior packages shall be packed in domestic-type exterior containers conforming to Specification PPP-B-585 (class 1), PPP-B-591 (style B), PPP-B-601 (style B), PPP-B-621 (class 1 — style 2), PPP-B-636 (type I or II — class 2), or MIL-B-10377 (style B). Closures shall be in accordance with the appendix of the applicable container specification. When fiberboard containers are utilized, the fiberboard shall have a minimum Mullen test of 275 pounds, and the gross weight shall not exceed the weight limitations of the applicable container specification.

5.2.3 *Level C.* Lights shall be packed in a manner which will insure acceptance and safe transportation at the lowest rate, to the point of delivery. The container shall comply with the Uniform Freight Classification Rules and regulations applicable to the mode of transportation which are in effect at time of shipment.

5.3 **Marking.** In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of Standard MIL-STD-129.

6. NOTES

6.1 **Intended use.** The portable light covered by this specification is intended for use to indicate a safe approach for an aircraft to a landing area.

6.2 **Ordering data.** Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) If batteries are to be furnished with the lights (see 3.5.1).

(c) If stage and clamp desired (see 3.5.8).

(d) Selection of applicable levels of packaging and packing (see Section 5).

6.3 **Drawing requirements.** The attention of contractors is invited to the fact that Government contracts for equipment, whether procured directly by the Government or through another contractor, require that all drawings submitted must be in accordance with the requirements of Specification MIL-D-70327, or other applicable drawing specifications.

6.4 **Battery dimensions.** Dimensions of military batteries may be appreciably greater than their commercial counterparts and contractors are cautioned to make ample provision for maximum dimensions permitted by the referenced battery specification.

6.5 **Provisions for preproduction tests.** Quantity production on a particular contract should be withheld until the preproduction sample has been pronounced satisfactory by the procuring activity. When a contractor is in continuous production of these lights from contract to contract, submission of additional preproduction samples on the subsequent contracts may be waived at the discretion of the procuring activity. Approval of preproduction samples or the waiving of preproduction tests does not preclude the requirements of acceptance testing.

6.5.1 It is understood that the lights supplied under contract will be identical in every respect to the corresponding approved preproduction sample and to the drawings except for changes which have been previously approved by the Government. Evidence of unauthorized change from the approved preproduction sample or drawings will constitute cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 202640-19T06:50Z
Check the source to verify that this is the current version before use.

Lights packaged as specified in 5.1 shall be packed in containers con

Notice. When Government drawings, specifications or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

Custodian:

Army—Mobility Command
Navy—Bureau of Naval Weapons

Preparing activity:

Navy—Bureau of Naval Weapons
Project No. 6210-0117

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:50Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 202610-19T06:50Z
Check the source to verify that this is the current version before use.